

LATINA, Italy

WMO: 162430

Lat: 41.5458N

Lon: 12.9097E

Elev: 26

StdP: 101.01

Time Zone: 1.00 (EUC)

Period: 90-02

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -0.2	(c) 0.9	(d) -6.1	(e) 2.3	(f) 6.0	(g) -4.0	(h) 2.7	(i) 5.3	(j) 8.8	(k) 12.5	(l) 7.6	(m) 11.2	(n) 0.8	(o) 320	(p) 0.369

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 10.6	(c) 32.8	(d) 23.7	(e) 31.2	(f) 23.6	(g) 30.1	(h) 23.2	(i) 26.0	(j) 29.8	(k) 25.1	(l) 29.1	(m) 24.3	(n) 28.3	(o) 3.6	(p) 210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.0	20.1	28.3	24.0	18.9	27.5	23.0	17.8	26.7	80.7	29.7	76.8	29.0	73.3	27.7	30.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 8.1	(b) 6.8	(c) 5.7	(e) -2.4	(f) 35.7	(g) 1.3	(h) 2.1	(i) -3.3	(j) 37.2	(k) -4.1	(l) 38.4	(m) -4.8	(n) 39.5	(o) -5.7	(p) 41.0		
(a) 8.1	(b) 6.8	(c) 5.7	(e) -3.2	(f) 27.8	(g) 1.2	(h) 1.3	(i) -4.0	(j) 28.7	(k) -4.7	(l) 29.4	(m) -5.4	(n) 30.1	(o) -6.3	(p) 31.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.8	8.4	8.5	11.1	12.9	17.6	21.4	24.2	25.0	20.9	17.6	12.8	8.9		
	DBStd	6.36	2.75	2.46	2.36	2.24	2.34	2.02	1.95	1.98	2.04	2.62	3.15	2.83		
	HDD10.0	198	64	51	15	3	0	0	0	0	0	0	9	56		
	HDD18.3	1524	309	274	223	164	42	2	0	0	3	47	167	292		
	CDD10.0	2324	13	9	51	89	236	343	439	465	328	234	94	22		
	CDD18.3	609	0	0	0	0	19	96	181	207	81	23	2	0		
	CDH23.3	5277	0	0	0	1	126	762	1784	2053	476	74	2	0		
	CDH26.7	1708	0	0	0	0	8	153	642	834	64	7	0	0		
(14) Wind	WSAvg	1.6	1.4	1.5	1.8	1.9	1.6	1.8	1.8	1.5	1.7	1.5	1.6	1.4		
(15) Precipitation	PrecAvg	969	104	80	82	79	50	36	17	46	86	120	150	111		
	PrecMax	1340	207	154	200	240	113	104	80	117	250	343	290	229		
	PrecMin	712	6	4	10	24	1	2	0	0	0	3	21	19		
	PrecStd	193	59	49	50	55	29	31	22	36	63	92	84	52		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.9	17.0	21.7	22.2	27.9	31.2	33.9	34.9	29.4	27.2	22.8	17.0		
		MCWB	13.8	12.2	13.7	16.5	19.6	22.8	23.7	24.1	21.2	19.7	19.3	14.4		
	2%	DB	15.2	15.8	19.1	20.9	26.0	29.8	32.2	33.1	28.2	25.2	20.2	15.9		
		MCWB	12.4	11.9	13.7	15.6	18.7	21.9	23.8	23.7	21.4	20.1	17.0	13.4		
	5%	DB	14.1	14.8	17.8	19.2	24.9	28.2	31.1	32.0	27.2	23.9	19.2	14.9		
		MCWB	11.6	11.2	13.2	14.9	18.8	21.3	23.5	23.6	21.3	20.1	16.6	12.8		
	10%	DB	13.1	13.8	16.1	18.1	23.2	27.1	30.0	30.9	26.1	22.7	18.1	13.9		
		MCWB	10.8	10.5	12.8	14.4	18.5	20.8	23.1	23.6	20.8	19.3	15.6	12.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.5	14.0	16.3	18.1	21.7	24.8	27.2	27.6	24.4	22.7	19.7	15.2		
		MCDB	16.1	15.8	18.8	20.5	25.2	28.6	30.6	30.8	26.6	24.5	21.5	16.4		
	2%	WB	13.2	12.7	15.0	16.6	20.6	23.7	25.9	26.4	23.5	21.7	18.2	14.2		
		MCDB	14.8	14.6	17.2	19.3	24.0	27.5	29.8	30.0	26.2	23.8	19.7	15.4		
	5%	WB	12.2	11.9	14.0	15.6	19.7	22.7	25.0	25.5	22.6	20.7	17.2	13.3		
		MCDB	13.8	13.8	16.5	18.4	23.2	26.4	29.2	29.4	25.6	22.7	18.9	14.5		
	10%	WB	11.2	11.1	13.2	14.9	19.0	21.9	24.1	24.7	21.7	19.6	15.9	12.4		
		MCDB	12.9	13.0	15.7	17.5	22.5	25.8	28.3	28.7	24.8	21.9	17.7	13.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.5	9.2	9.4	9.1	9.8	10.4	10.7	10.6	9.4	8.6	7.9	7.9		
		MCDBR	8.6	10.3	11.4	11.0	11.7	12.1	11.9	12.5	10.6	9.6	7.8	8.0		
		MCWBR	6.3	7.3	7.3	7.2	6.4	6.5	5.9	5.9	5.9	6.1	5.5	6.1		
	5% WB	MCDBR	7.8	8.8	9.9	9.7	10.3	10.6	10.7	10.8	9.2	8.2	6.9	6.8		
		MCWBR	6.2	7.1	7.1	7.0	6.6	6.7	6.3	6.4	6.0	6.0	5.4	5.9		
(40) Clear-Sky Solar Irradiance	taub		0.354	0.380	0.422	0.452	0.456	0.449	0.441	0.441	0.432	0.416	0.390	0.355		
	taud		2.395	2.295	2.175	2.117	2.156	2.230	2.258	2.269	2.282	2.290	2.292	2.385		
	Ebn at Noon		782	814	817	820	826	831	833	822	800	763	723	744		
	Edh at Noon		83	106	133	151	149	139	134	128	119	105	91	78		
(44) All-Sky Solar Radiation	RadAvg		1.77	2.65	3.88	5.17	6.30	7.19	7.37	6.44	4.69	3.24	2.00	1.61		
	RadStd		0.17	0.27	0.45	0.34	0.51	0.32	0.36	0.37	0.23	0.28	0.19	0.18		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (16 neighbors)	+0.41	N/A	N/A	+0.53	N/A	N/A	N/A	-79	+131	+68

Nomenclature: See separate page